

Project Documentation

EV MARKET ANALYSIS – 2026

1. Project Overview

The EV Market Analysis 2026 project focuses on performing an end-to-end data analytics workflow using Python in Jupyter Notebook to analyze electric vehicle market data. The project involves data collection, preprocessing, cleaning, exploratory data analysis (EDA), and visualization to uncover meaningful insights from the dataset.

The analysis examines various aspects of the EV market, including vehicle brands, models, pricing, battery capacity, driving range, drive types, customer ratings, and annual sales performance. Through statistical analysis and visualizations, the project identifies market trends, customer preferences, and relationships among key variables.

Various data analysis techniques such as filtering, sorting, grouping, pivot tables, cross-tabulation, and data visualization are used to explore the dataset and support data-driven decision-making. The findings provide a better understanding of the current EV market landscape and help highlight factors influencing vehicle performance and market success.

Overall, this project demonstrates how data analytics can be applied to transform raw EV market data into valuable business insights, enabling a deeper understanding of industry trends and consumer behavior.

2. Tools Used

Python

- Primary programming language used for data analysis and visualization.
- Provides powerful libraries for data manipulation and analytics.

Jupyter Notebook

- Interactive development environment used to write, execute, and document Python code.
- Helps combine code, outputs, visualizations, and explanations in a single notebook.

Pandas

- Used for data loading, cleaning, preprocessing, and analysis.
- Functions used include:
 - `head()`
 - `info()`
 - `describe()`
 - `groupby()`
 - `pivot_table()`
 - `crosstab()`
 - `sort_values()`
 - `value_counts()`

NumPy

- Used for numerical computations and handling missing values.
- Supports mathematical and statistical operations.

Matplotlib

- Used for creating basic visualizations such as:
 - Bar Charts
 - Histograms
 - Scatter Plots
 - Pie Charts
 - Subplots

Seaborn

- Used for advanced statistical visualizations.
- Provides attractive and informative plots such as:
 - Box Plots
 - Heatmaps
 - Count Plots

Plotly

- Used for interactive visualizations.
- Enables dynamic charts and dashboards for better data exploration.

Exploratory Data Analysis (EDA) Techniques

- Univariate Analysis
- Bivariate Analysis
- Multivariate Analysis

Data Visualization Techniques

- Bar Charts
- Line Charts
- Scatter Plots
- Histograms
- Pie Charts
- Box Plots
- Heatmaps
- Subplots

3. Dataset

- Source : Sample data
- Data contains : Brand , Model , Variant , Body_type , Drive_type , Battery_capacity_kwh , Range_miles , Price_usd , Customer_rating , Annual_sales_units , Market_segment , Country_of_origin etc.

4. Steps Followed

1. Data Collection
 - Collected EV market 2026 data from CSV file.
2. Data Cleaning and Preprocessing (Pandas , NumPy)
 - Handling Missing Values
 - Removing Duplicate Records
 - Correcting Data Types
 - Standardizing Data
 - Creating Derived Columns
 - Filtering Unnecessary Data
 - Validating Data Quality
3. Exploratory Data Analysis (EDA)
 - Univariate Analysis
 - Bivariate Analysis
 - Multivariate Analysis
4. Data Visualization (Matplotlib , Seaborn , Plotly)
 - Bar Chart
 - Line Chart
 - Scatter Chart
 - Histogram
 - Pie Chart
 - Box Plot
 - Subplot
 - Count Plot

5. Key Insights

- **EV Market Diversity** - The dataset contains a wide range of electric vehicle brands, models, body types, and drive types, reflecting the growing diversity of the EV market.
- **Price and Range Relationship** - A positive relationship exists between vehicle price and driving range, indicating that higher-priced EVs generally offer longer battery range and enhanced performance.
- **Battery Capacity Impact** - Vehicles with larger battery capacities tend to provide greater driving range, highlighting the importance of battery technology in EV performance.
- **Sales Performance** - Certain EV brands significantly outperform others in annual sales, indicating strong market presence and customer preference.
- **Drive Type Distribution** - AWD, FWD, and RWD vehicles are distributed across different market segments, with AWD models often associated with premium and high-performance EVs.
- **Customer Satisfaction** - Most vehicles received relatively high customer ratings,

suggesting positive consumer acceptance and satisfaction with electric vehicles.

- **Market Segment Trends** - SUVs and premium EVs represent a significant portion of the market, indicating increasing demand for spacious and feature-rich electric vehicles.
- **Brand Comparison** - Some brands offer higher average range and customer ratings, while others focus on affordability and market accessibility.
- **Overall Market Growth** - The analysis indicates a rapidly expanding EV market driven by technological advancements, increasing consumer adoption, and growing competition among manufacturers.

6. Screenshot



7. Files Included

- 'ev_market_2026.csv' – Uncleaned data
- 'EV Market 2026_Final.ipynb' – Cleaned Visualized file
- 'README.md' – Project description

8. How to Use

- Open `EV Market 2026\_Final.ipynb` to view the full steps of the project.